

Textile & Fiber

Classification of Synthetic fiber

Fiber :-

A fiber is a thin, thread like substance that is significantly longer than its wide. Fibers can be natural or man-made (synthetic) and are classified on the base of their origin.

Classification of fiber:-

1. Natural Fiber:

These fibers are derived from natural resources e.g: plants, animals & minerals.

• Vegetable fiber:

These come from plants & are mainly composed of cellulose e.g: cotton, flax (linen), jute, hemp & sisal.

• Animal fiber:

These are derived from animals & are primarily made of protein.

e.g: wool, silk, mohair.

• Mineral fiber:

These are inorganic materials that are shaped into fibers

e.g: asbestose.

2. Synthetic fiber:

These are produced artificially or through modification of natural material.

- Regenerated fiber:

These are made from natural polymers like cellulose, such as rayon & lyocell.

- Synthetic fiber:

These are made from synthetic polymers like nylon, polyester and acrylic.

- Inorganic fibers:

These are made from inorganic materials like glass and carbon fibers.

Classification of synthetic fiber

(A) Based on chemical composition / Polymer Type.

This is the most common and fundamental way to classify synthetic fibers.

i) Polyamides (Nylon): 1st truly synthetic fiber by strong amide linkage ($-CONH-$). They are strong, durable, elastic, abrasion-resistant, quick drying, resistant to insects, fungi and many chemicals.

e.g: Nylon 6, Nylon 66.

Uses: Hosiery, lingerie, swimwear, ropes, parachutes, Carpets, automotive parts, fishing lines, industrial fabrics.

ii) Polyester:-

The fibres which are composed of ester functional group ($-\text{COO}-$). The most common type is polyethylene terephthalate (PET).
→ Polyester is excellent wrinkle resistance, high strength, good dimensional stability, resistance to stretching & shrinking, quick-drying, resistant to most chemicals, mildew and abrasion. It is thermoplastic.

→ Uses:

Used in home furnishings (sheets, curtains) blankets, carpets, industrial fabrics (belts, hoses), plastic bottles, films.

e.g. Polybutylene terephthalate, PCT (polycyclohexylene dimethylene terephthalate)

iii- Polyolefins:-

The fibres which are made from polymers of olefins, primarily propylene and ethylene.

Commonly polypropylene & polyethylene is used.

Properties:

- Very light weight
- excellent moisture-wicking
- good insulator
- High strength to weight ratio
- Chemical resistance
- low melting point as compared to other synthetics.

Uses:

- Athlete wear
- Carpets
- Outdoor gear
- geotextiles
- Ropes
- disposable medical garments.

N- nylons:

These fibers are composed of at least 85% by weight of acrylonitrile units, commonly polyacrylonitrile (PAN) is used.

properties:

- soft, warm, light weight,
- wool-like, good bulk
- resistant to moths, oils & most chemicals
- good sunlight resistance
- moderate strength & abrasion resistance.

Uses:

It is used in sweaters, tracksuits, blankets, carpets, upholstery, craft yarns, outdoor fabric.

v- Elastomeric fibers:

Synthetic fibers are known for their elasticity & ability to stretch significantly and return to their original shape.

Spandex (known as Lycra in many regions or Elastene in Europe) a commonly

used type of elastomeric fiber.

properties:

- High stretch upto 500-700%.
- excellent recovery
- Resistant to body oils & perspiration.

Uses:-

- Blended with other fibers to add stretch to clothing (leggings, swimwear, activewear, hosiery, denim).

vi- Aramids:

These are the high performance polyamides characterized by aromatic rings in their polymer backbone, providing exceptional strength & heat resistance.

Para-aramids and meta aramids are common aramids used as fibers.

Properties:

- Very high strength to weight ratio
- High thermal stability (don't melt)
- flame resistance
- impact resistance
- chemical resistance

Uses:

- Bulletproof vests, military helmets, protective gloves, ropes, aerospace components, tires, fiber optics.
- Firefighter uniform, industrial protective clothing, electrical insulation, hot gas filtration.

vii- Polyvinyl Derivatives:

These ^{which} fibers are derived from polyvinyl chloride (PVC) or polyvinyl alcohol (PVA), are vinylon and vinylon respectively.

Properties:

- Vinylon: Resistant to water, chemicals, & non-flammable. low softening point.

→ Vinyon: High strength, good chemical resistant, absorbant (unlike others)

Uses:

Vinyon: Filters, industrial fabrics, fishnets, outdoor awnings

Vinyon: Workwear, fishing nets, ropes, Canvas

viii. Others:

• Modacrylic:

Modified acrylics with enhanced flame resistance, Used in protective clothing, wigs, upholstery.

• Carbon Fiber:

Extremely strong, lightweight, rigid, made from carbonized organic polymers, Used in composites for aerospace, sports equipments, automotive.

• Glass Fiber:

Strong rigid, non-combustible, excellent electrical ~~conductor~~ insulator, reinforcement in plastics, optical fiber.

• PBI: (Polybenzimidazole fiber)

Exceptional thermal and chemical stability, often used in high performance protective apparel.

(B) Classification Based on processing method:

While primarily classified by chemical type, processing method leads to variations.

• Melt spun fibers:

Polymers are melted & extruded through spinnerets e.g. (Nylon, polyester, polypropylene)

• Dry spun fibers:

Polymer is dissolved in solvent & the solvent is evaporated e.g. (acrylics, spandex)

• Wet spun fibers:

Polymer is dissolved in a solvent and extruded into a coagulation bath e.g. some acrylics

(C) Semi-Synthetic fibers:

They are derived from natural fiber polymers but undergo significant chemical process.

• Rayon (Viscose):

It is soft, absorbent, comfortable, drapes well, mimics properties of silk & cotton.

It is regenerated from cellulose (from wood pulp)

It is used in clothing (dresses), linings, upholstery, bedsheets, carpets.

• Acetate:

It is originated from cellulose esters, it is soft, shiny, drapes well. Used in Dressmaking, linings.

(D) Classification Based on Behaviour during heating:

• Thermoplastic fibers:

These fibers melt or fused when heated appropriately & can often be dissolved in specific solvents. Most synthetic fibers fall into this category.

Example: Nylon, polyester, PVC, cellulose acetate.

• Non-Thermoplastic fibers:

These fibers tend to decompose on heating before exhibiting plastic flow & are rarely soluble in solvents.

example:

Some specialized high performance fibers.

Chemistry & manufacturing of Viscos Rayon:-

Rayon is a semi-synthetic fiber made from regenerated cellulose. This means it starts with a natural polymer (cellulose, typically from wood pulp) but undergoes significant chemical processes to transform it into a usable fiber.

Raw materials:-

The 1937, consumption of the main raw materials in the manufacture of rayon, & its allied industry, the manufacture of cellulose film is:

Sulfuric acid

Acetic acid

Caustic soda

Soda ash

Sodium sulfates

Ammonia

Glucose

Cotton linters

Wood pulp

Carbon bisulfide

Upto a few years ago, the cuprammonium and acetate processes used only cotton linters as their source of cellulose.

Rayon / Artificial Silk

Originally rayon was called the artificial silk but now all the artificial fibers obtained from cellulose are given the name Rayon.

Artificial silk is different from natural silk. The natural silk is proteinous material, whereas rayon is carbohydrate in its origin.

These are two types of ~~cotton~~ fiber.

i- Regenerated fibers.

ii- Cellulose fibers.

i- Regenerated fibers.

These fibers are similar to the original cotton. These are made from polymers.

ii- Cellulose derivatives:

The cellulose derivatives have properties similar to plastic from which they are made.

Types of Rayon:

i- Viscose rayon

ii- Cuprammonium rayon

iii- Cellulose acetate rayon

iv- Nitrocellulose rayon or pyroxylin

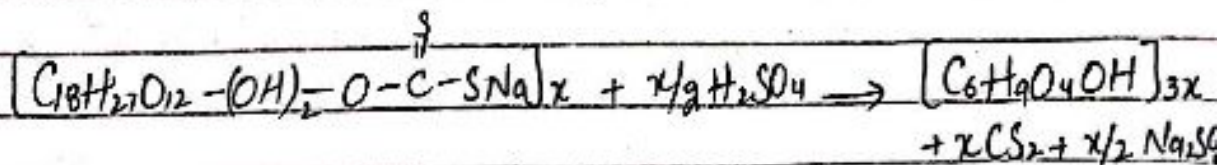
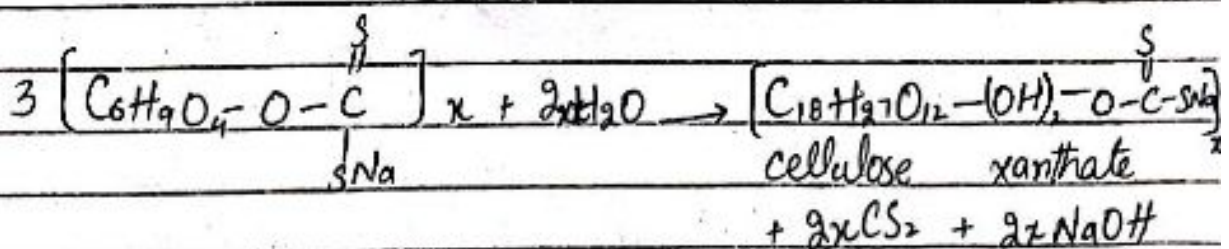
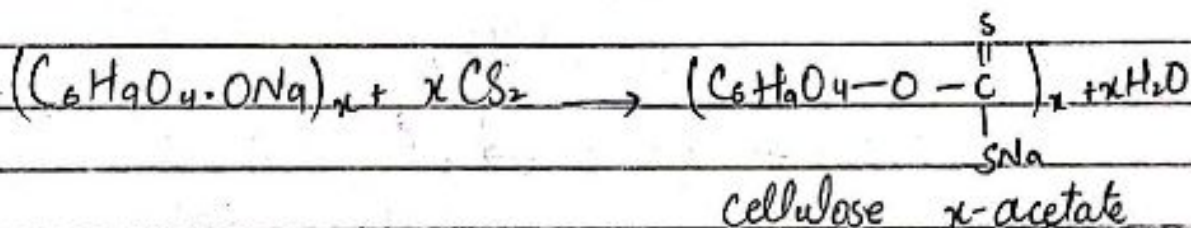
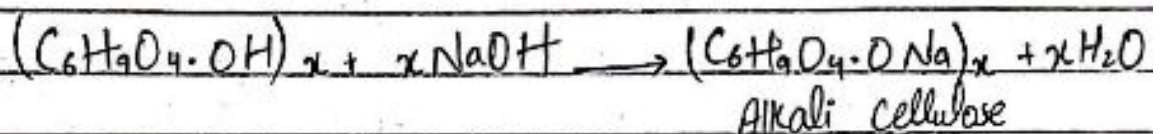
i- Viscose Rayon:

80% of the world production of artificial silk is got by this process. It is made from wood pulp and wood pulp + cotton linter pulp containing excess proportion of wood pulp.

It is very useful method for the production of regenerative type of rayon.
Raw Material:

- Cellulose $\Rightarrow$ cotton linter + wood pulp
- acetic anhydride, NaOH
- acetic acid, Cl_2 /hypochloride

Chemical reactions:



The finished product is pure cellulose but it contains smaller molecules of original wood pulp or cotton.

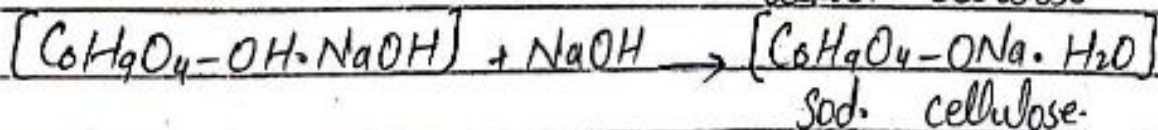
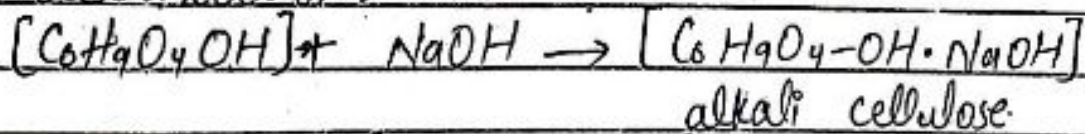
Rayon Manufacturing

Viscose rayon formation is carried out in the following steps.

1- Alkali Cellulose formation:

Cellulose is 1st bleached and then charged into vertical perforated sheets plates in 18% NaOH for about 4 hours at about 56-62 °F. The cellulose still contain some hemicellulose in the excess layer which is drained off and recycled.

The partial destruction of intermolecular bonds, penetration of NaOH in swollen amorphous region of cellulose. Where it is held by H-bonding & sodium. cellulose is formed, is called mercerization.



Shredding:-

The soft sheets of sod. cellulose are reduced to smaller crumbs (fluffy white flakes) in a shredder surfactants are added.

Time = 2-3 hours

Temp = 65-88 °F

Ageing or pre ripening:

The crumbs

are aged for 48-72 hrs. at 75°F in steel cans in the presence of air. Aging or pre-ripening cause some oxidation and degradation.

ii. Formation of Cellulose xanthate:

The alkali cellulose crumbs thus formed are separated and taken in a horizontal iron drum called Bassette, which moves with narrow axis. This alkali cellulose is treated with 60% of its weight with CS_2 . The drum is rotated during the addition of CS_2 & Temp. is maintained at 20°C for 3 hours.

The crumbs first turn yellow than deep orange and co-agulated into small balls of cellulose xanthate.

The composition of cellulose xanthate depends upon the amount of CS_2 used, duration of xanthation period & Temperature.

iii. Formation of Viscose solution:

Xanthate is soluble in NaOH , giving rise to ~~viscous~~ viscous soln. known as viscose.

The cellulose xanthate balls are chopped

into 3-4% NaOH soln. in a dissolved. The final product containing 6% cellulose & 8% NaOH is obtained. The colour & odour is due to Na_2CO_3 & H_2S .

If desired some delustering agent (TiO_2) or organic pigments are added to viscose soln. in the mixer. Several batches of viscose solution are blended and refined for 4-8 days under controlled conditions at 60% till the cellulose approaching nearly to coagulating point.

The %age of combined sulfur decreases somewhat and some xanthic acid splits off and a part of cellulose is regenerated.

iv. Ripening Period:

During ripening period, solution is filtered several times, through cotton pads to remove dust, dirt and insoluble material so that it should not clog spinnerets.

Ripening is necessary to impart strength & spinning properties in the yarn.

Finally the soln. is placed under to remove bubbles which may break the continuity of the filament.

v- Spinning Machine:

The soln. is fed into the spinning machine.

2 Types of machines are used.

i- Bucket Machine

ii- Bobbin Machine

The soln. is extruded under pressure through the spinnerete into the spinning bath. The spinneretes are small caps of noble metal with holes through which the soln. is extruded.

Just at the head of spinnerete there are filters to remove any solid particles that might clog the hole.

The spinning soln. contain 8-10% H_2SO_4 (To neutralize $NaOH$).

13.5 - 21.5% $NaSO_4$.

1% $ZnSO_4$

4-10% Glucose (To prevent crystallization).

The soln. from the spinnerete is coagulates in the spinning bathes as filament of regenerated cellulose.

i- Bucket Type:

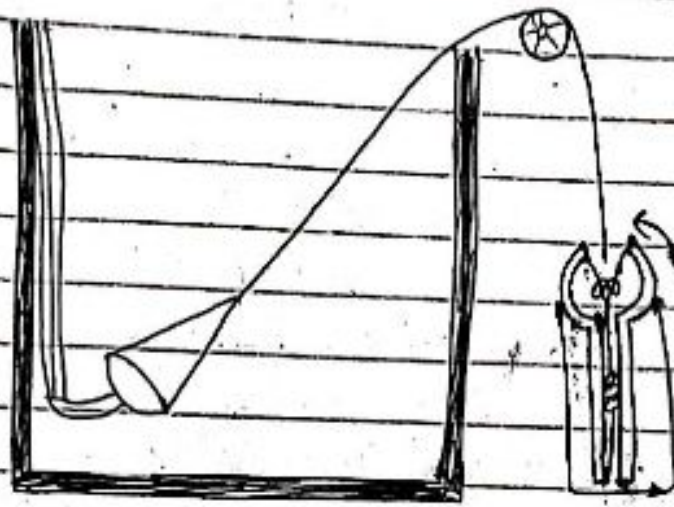
If bucket machine is used the spinnerete head dipped horizontally in soln. bath.

Several filaments gathered into thread. The thread is fed into a

small centrifugal bucket revolving at 7500 rpm the machine.

(a) Twist the thread

(b) Remove a greater portion of the liquor through the proportion in the periphery.



Bucket spinning.

ii- Bobbin machine:

In bobbin machine, the spinnerette head is vertically upward into spinning bath. The filaments are wound on a revolving bobbin. There is no twist imparted to thread in bobbin.

Yarn from either type is washed to remove the spinning liquor.

The bobbin yarn is dried, twisted and skeined.

The bucket yarn is dried

The bucket yarn is skeined without dyeing

Both types of yarn are desulphurized by treating with $3/4 - 1\%$ Na_2S soln.

Finishing:

Both types of yarn are washed, bleached, washed again and coned.

Importance:

This is the cheapest and commonly used process as it provides cheap wood cellulose.

While others require costly or poor materials as in cuprammonium and nitrocellulose process.

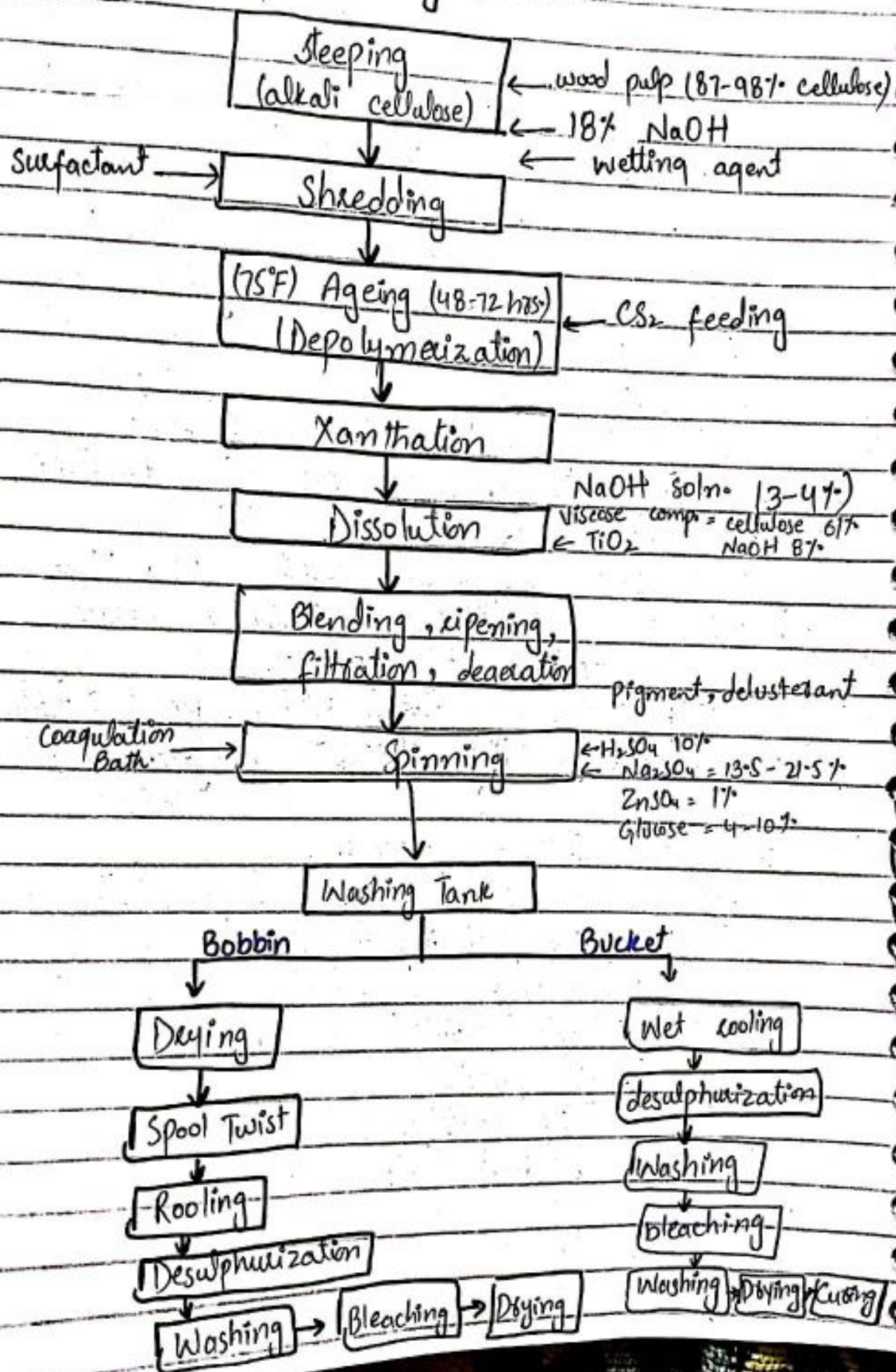
In USA 10% of total yarn production is prepared by viscose rayon.

Uses of Viscose Rayon:

Viscose rayon are widely used in making

- Dress linings
- Curtains
- Table cloths
- Bed sheets
- Fine fabrics
- Cushions
- Sports wear.

Flow sheet of Manufacturing of Viscose Rayon:



Properties of Viscose Rayon

Specific gravity:

Higher than wood & cellulose acetate. It is about 1.52.

Action of Heat:

Long exposure to heat cause decomposition.

Action of acids:

Not very resistant, attacked by acids at 55°C or higher & undergo carbonization.

Action of solvent:

Dry clean solv^s have no action.

Action of light:

More resistant than real silk, but weakened by continuous exposure.

Moisture content: 10-12% under standard conditions.

Creep

When stretched for sometime & then released, it does not readily come to its original position, it takes sometime. This is called delayed elasticity or Creep of Viscose.

Moisture absorption:

Have high moisture.

Insulating properties:

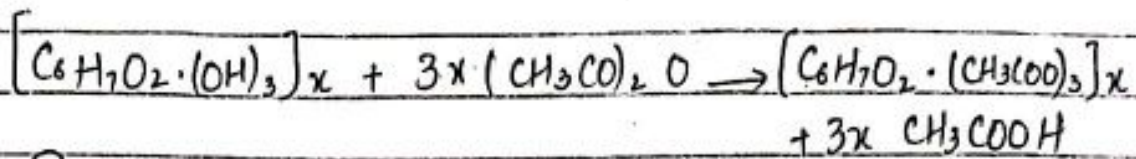
Not good insulator due to high moisture content.

Cellulose acetate OR

Acetate Rayon

The cellulose acetate is not regenerative cellulose ester of cellulose.

Reaction:-



Raw materials:-

acetic anhydride $(CH_3CO)_2O$

Glacial acetic acid (CH_3COOH)

H_2SO_4

Cellulose (cotton linter)

Manufacturing of acetate Rayon:-

⇒ Formation of Triacetate Ester:

100lb of CH_3COOH
+ 100lb of $(CH_3CO)_2O$ + small amount of H_2SO_4
as catalyst are added in a jacketed glass,
lined with cast iron. The mixture is cooked
in an aging tank at $45^\circ F$ and 35 lb.
Cotton linters are added slowly with
constant stirring.

The acetylation requires 5-8 hrs. at
temp. $286^\circ F$.

The viscose soln. is diluted
with equal parts of conc. CH_3COOH and
10% H_2SO_4 and allowed to age for

15 hrs. at 63°C .

The hydration of acetate group starts. It is necessary to transform the cellulose into triacetate (because cellulose is not directly converted into desired acetyl group). and partially hydrolyze the required production of acetate groups, the water formed is taken up by acetic anhydride & converted into acetic acid.

Formation of diacetate ester:

The hydration is stopped by the running of mixture into a large volume of water and preparing the secondary acetate.

The triacetate is soluble in chlorinated mixture and gives brittle threads so this is reason why it is converted into diacetate form.

The sec. acetate is centrifuged to separate it from acetic acid, which is recovered, then concentrate it for further use.

Precipitation:

The aged soln. is then mixed with an excess of water causing ~~like~~

the cellulose acetate to precipitate as white flakes. The flakes are washed several times by decantation and then centrifuged to remove the remaining liquor.

Formation of spinning soln:

The flakes are dried, ready for preparing spinning soln. The spinning soln is made by dissolving dry flakes in acetone in an agitated mixer.

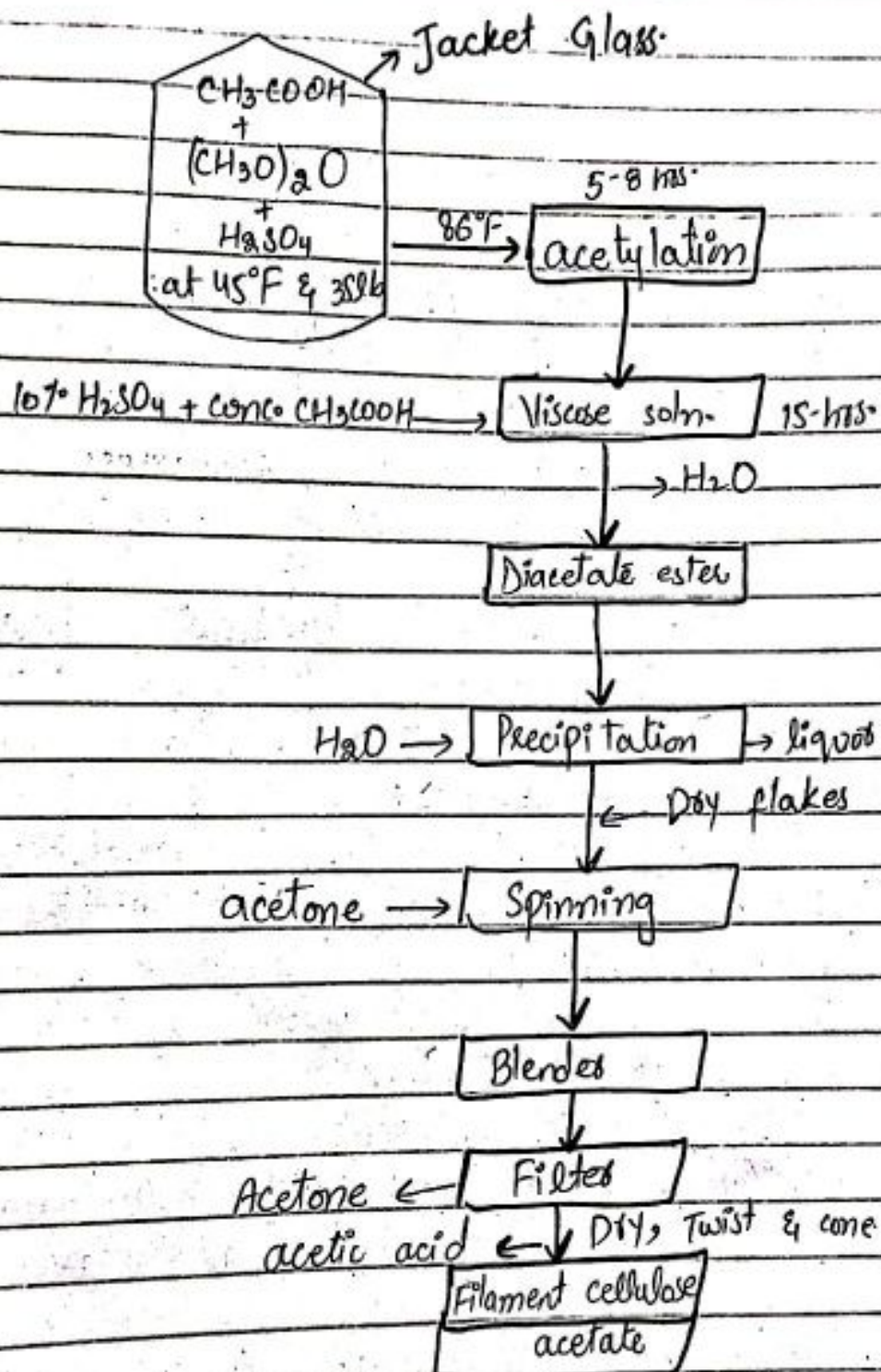
If desired, the delustering material may be added.

Spinning Machine:

Several batches are blended, filtered and sent to the spinning machine.

The solution is forced through the spinners to a current of warm mostly air. Acetone separates and is recovered leaving a filament of cellulose acetate. The filaments are dried, twisted and coned in the same manner like the viscose rayon or Cuprammonium. Acetic acid is recovered & concentrated.

Flow sheet



Uses of acetate Rayon

- It is soft & produce warm feelings.
- It is used for all kinds of mens & women wear such as bies, pajamas, dressing, gowns, socks, undergarments, bathing & swimming suits etc.
- It is also used as insulating wear for electrical wires & coils due to low hygroscopicity it is not suitable for making handkerchiefs. & ~~fabrics~~.

Nylons

The word Nylon is general name. Now it is used for a no. of comp. as Nylon 6, Nylon 66, Nylon 610 having amide $(-NHCO-)$ linkage.

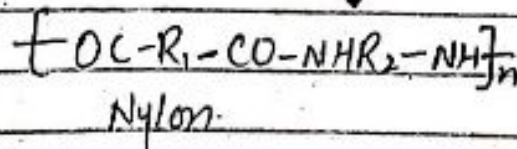
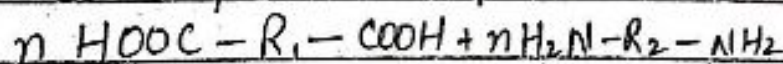
Nylon was the product of Wallace H.C. and his co-workers in 1921.

Wallace H.C. 1st time introduced Nylon from adipic acid & hexamethylene di-amine with molecular weight ranging from 2500-5000.

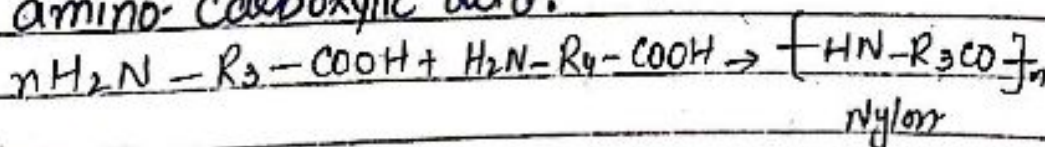
Preparation:

Nylons are in general obtained by the condensation polymerization reaction of either a Dicarboxylic acid & a diamino acid and aminocarboxylic acid themselves.

i- Di-carboxylic acid; & Di-amine:



ii- amino-carboxylic acid:

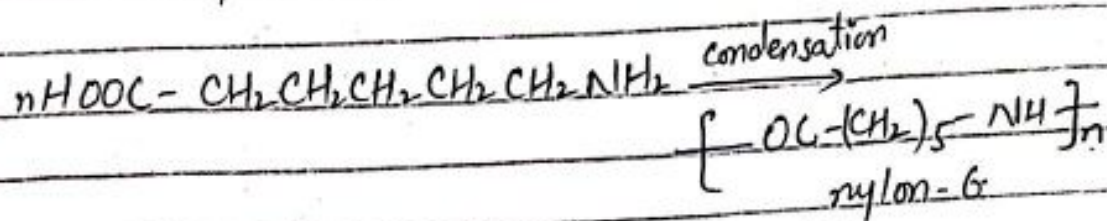
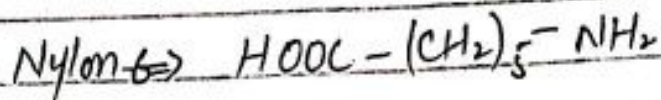


The value of R group depends upon the nature of nylon to be formed.

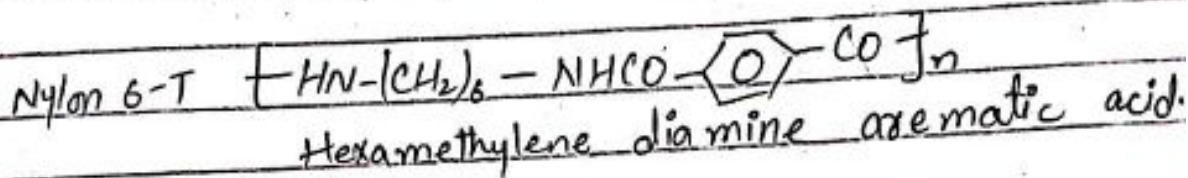
Types of Nylon

Nylon 6 preparation:

Nylon-6 is formed by the condensation reaction of amino-caproic acid.

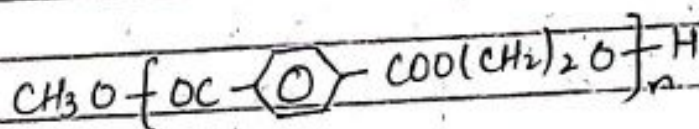


Aromatic polyamide: Nylon 6-T:-

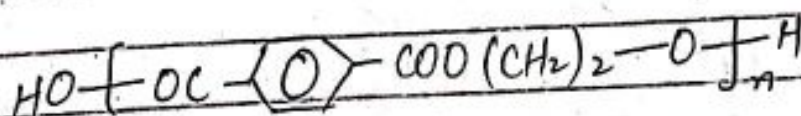


Polyesters:-

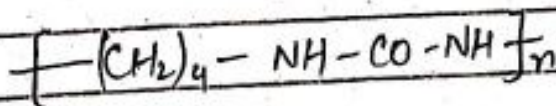
i- Terylene:



ii- Dacron:



Poly Urea:

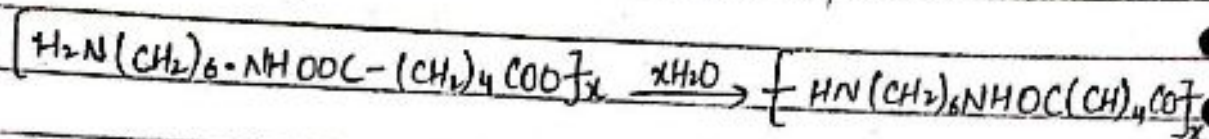


Manufacturing of nylon:

Both adipic acid & hexamethylene diamine are soluble in water and both are mixed in 1:1 ratio in an autoclave & heated at about 100°C . As a result of reaction Nylon salt is formed in solution.

$$n\text{HOOC}(\text{CH}_2)_4\text{COOH} + \text{H}_2\text{N}(\text{CH}_2)_6\text{NH}_2 \longrightarrow \left[\text{H}_2\text{N}(\text{CH}_2)_6 \cdot \text{NHOC}(\text{CH}_2)_4\text{COO} \right]_n$$

Nylon salt.



Polyhexamethylene Adipamide or Nylon salt is transferred into an evaporator to remove water molecules. Some acetic acid is added as stabilizer to stabilize the viscosity of the soln. After evaporation, salt is converted into Nylon 66 by heat.

The salt is transferred to autoclave under N_2 . After evaporation the salt soln flows into jacketed autoclave equipped with internal heated coils. Here's rest of water is removed.

The pigment dispersion agent TiO_2 is added & polymerization takes place.

After completing polymerization the molten syrup polymer is forced out of the bottom into a casting wheel by using specially purified N_2 at 40-50 lb.

A ribbon of polymer about 17ft wide & $1/4$ inch thick flows into 6ft

Casting drum. Water spray on the inner side cool and harden the ribbon. While outside is cooled by air.

The ribbons are cut down into small chips or flakes before blended uniformly. Two or more batches are mixed in the light blenders. These blenders empty into hoppers from where the flakes are transferred to spinning area.

Special precautions are made to keep the whole process & spinning operation on fire. The spinning unit is composed of a metal vessel surrounded by a down turn heated jacket which keep the temp above M.P. (263°C) In vessels, the flakes strikes the grid & melt in melting chamber.

Molten polymer flows through the ports in this chamber to gear spinning pump. The polymer is passed through the filter screens & then through spinners, the ~~sp~~ filaments are solidified by air in a cooling chimney.

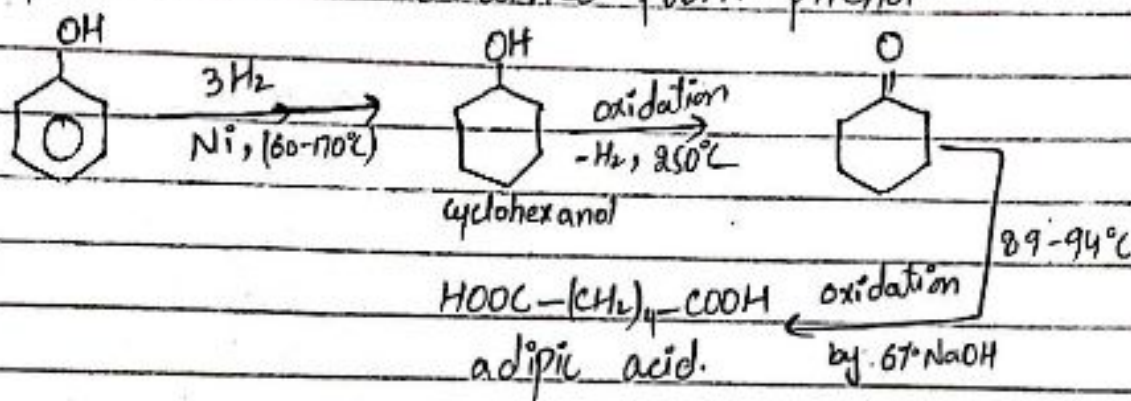
The fibers are passed through a stream of humidified air to prevent brittleness.

The fibers are lubricated in glass rolls, to give finishing. The filament is then stretched to the desired degree by passing through a roller system. Here the strength & elasticity characteristics of yarn are developed.

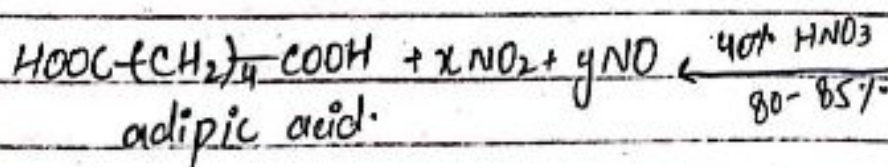
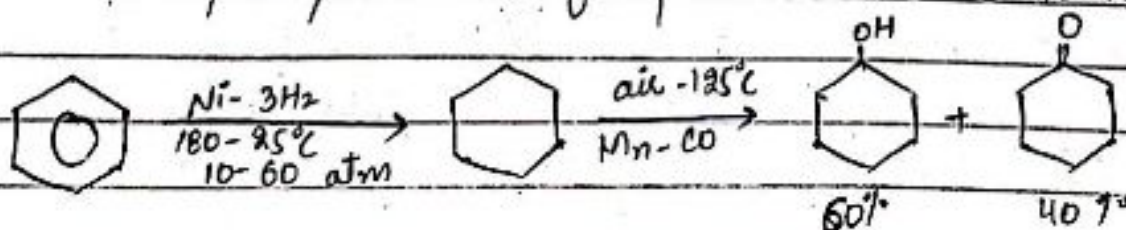
The resulted filament is coiled & shipped.

Preparation of Nylon Intermediates: by adipic acid:

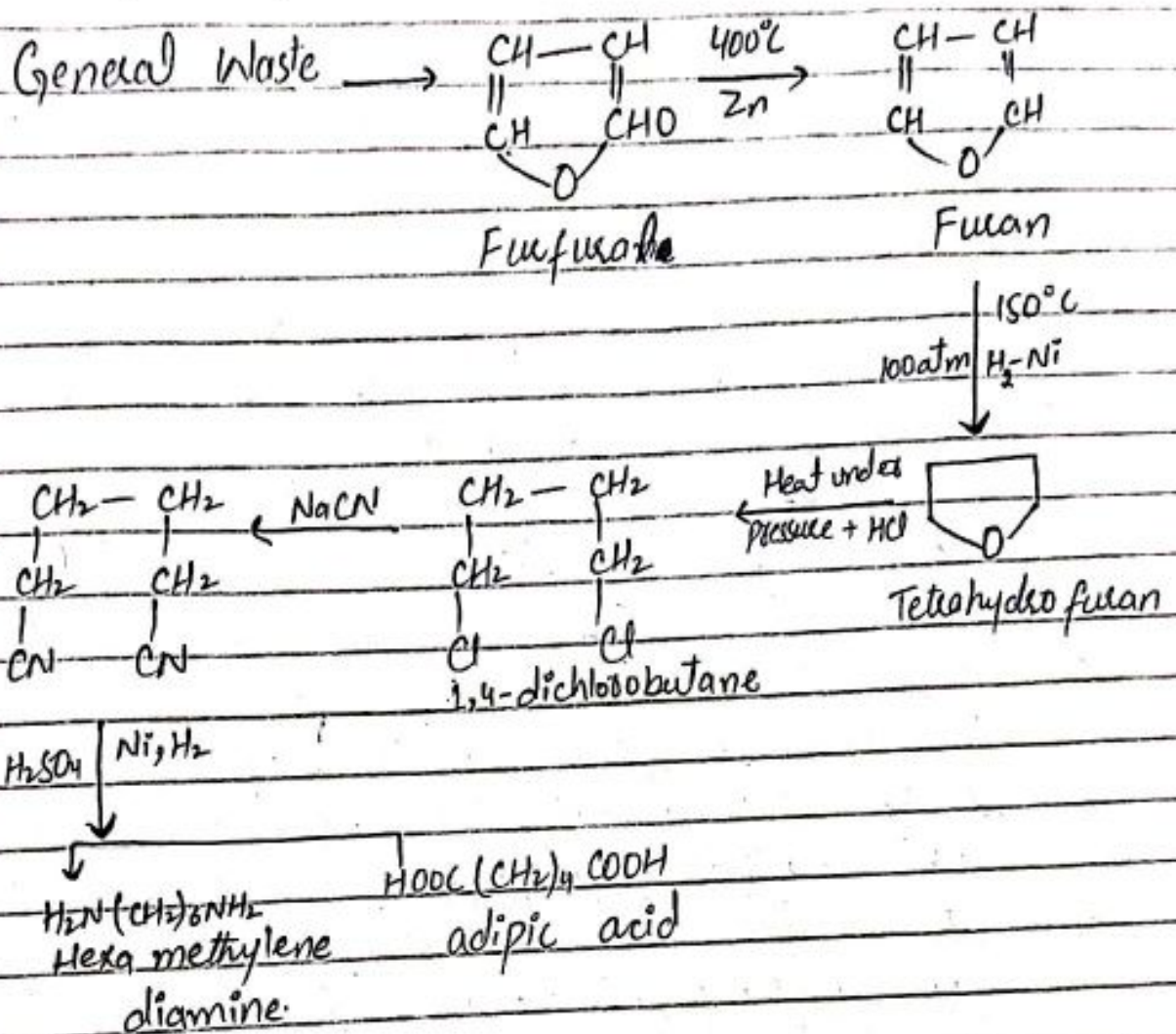
Adipic acid was first produced on large scale by the oxidation of cyclohexanone obtained from phenol.



Now adipic acid is largely produced by the oxidation of cyclohexane obtained by the hydrogenation of petroleum or benzene.



Nylon from Cereal Waste : (Husk)



Properties of Nylon 66 :-

specific gravity	: 1.14
M.P	: 263°C in N_2 & 250° in air
Moisture resistance	: very good
Elasticity	: High upto 150°C
Toughness	: High upto 150°C
Strength & abrasion resistance	: High
Solvent resistance	: Good
Action of air & water	: attacked at elevated temp.

Uses of Nylon:

⇒ Nylon is suitably used for fabric sewing thread & heavy rope due to its high tenacity.

⇒ Because of much longer life nylon bonded rubber tyres are very popular in auto-mobile & aircraft.

⇒ Nylon is being used in making transmission belting, link belts, ribbons for elastic type, spind bands on textile spinning machine.

⇒ Nylon is used in textile fibers.

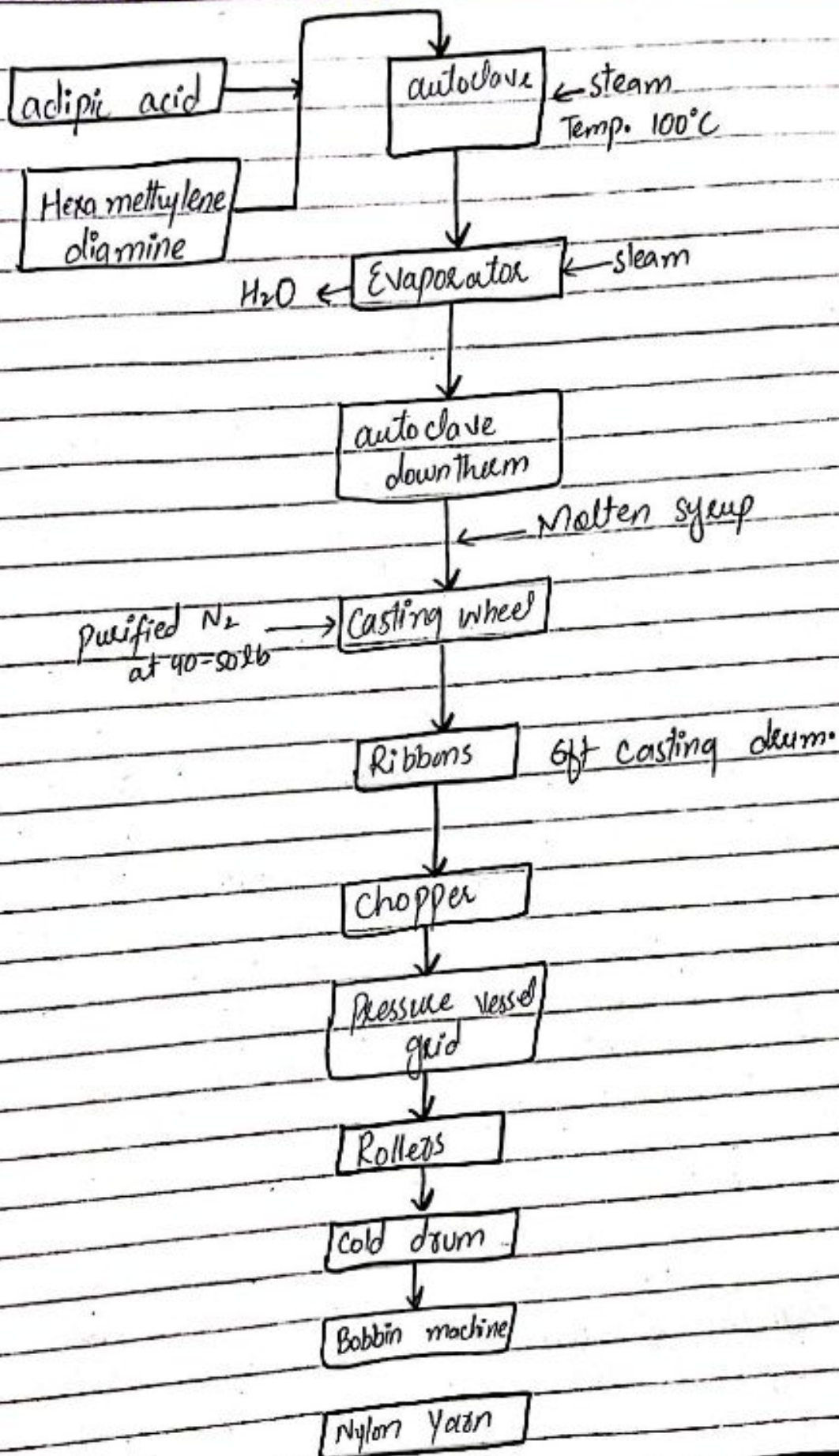
⇒ Nylon is used in carpet industry.

⇒ Nylon insect screens are popular due to their biological resistance.

⇒ It is used in oil filters due to its chemical stability.

⇒ It is also used in pottery industry in making filter sacs.

Flow sheet:-



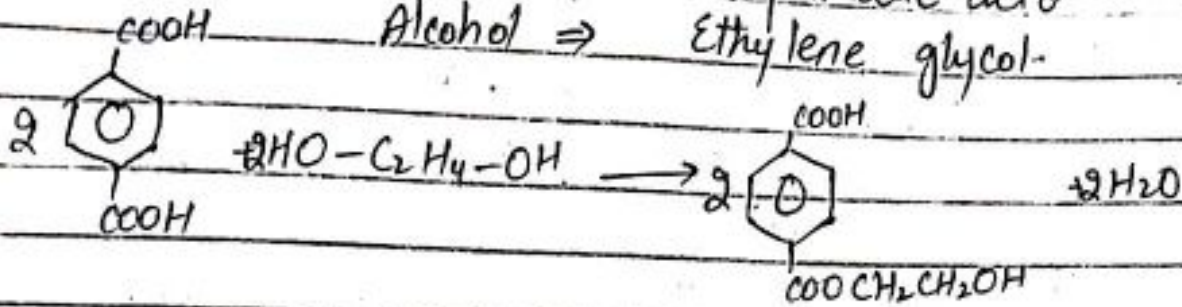
Polyesters:-

TERYLENE OR Dacron:

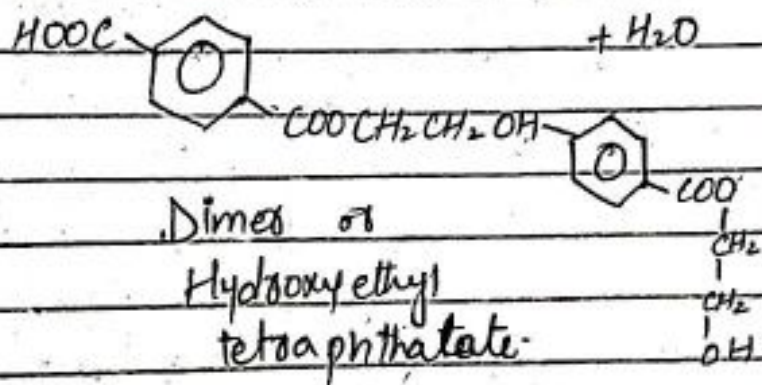
ester prepared from acid & alcohol. It is a polymeric

Acid $\Rightarrow$ Terephthalic acid

Alcohol $\Rightarrow$ Ethylene glycol.



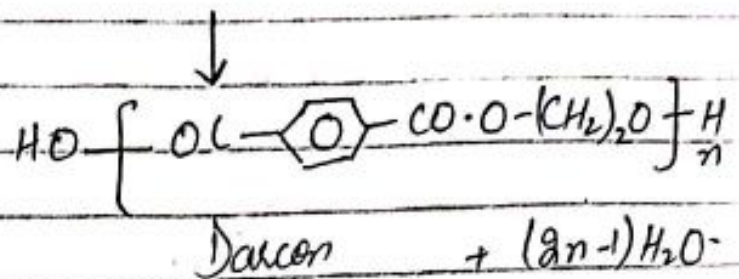
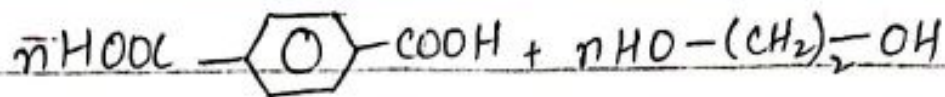
hydroxy ethyl terephthalate



The dimer still contains reactive group at either ends so that polymerization proceeds. A high polymer comprising to benzene nuclei is enough for good fiber.

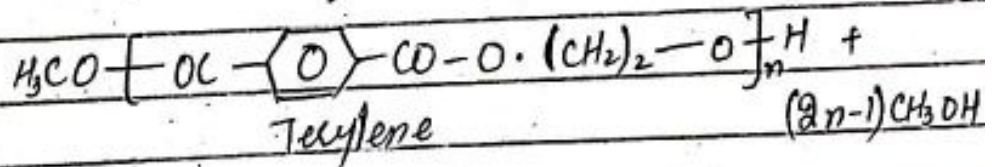
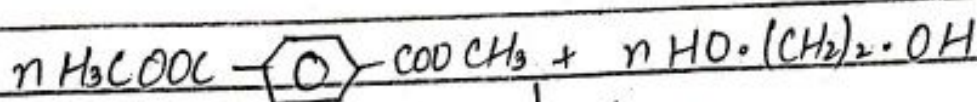
Dacron:

In ~~am~~ America, polyester is prepared from terephthalic acid and called dacron.



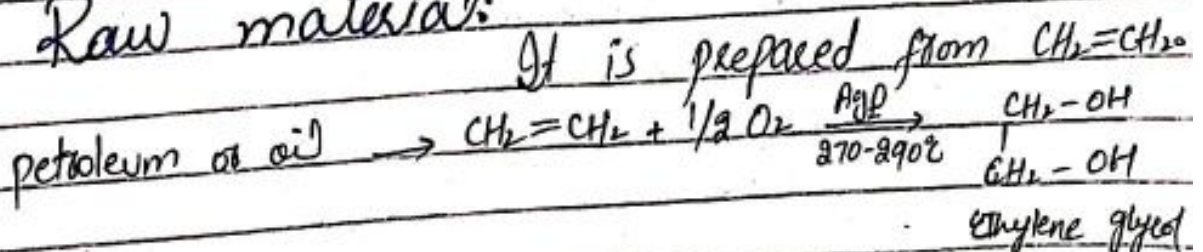
TERYLENE:-

The british fiber terylene is made by the polymerization of dimethyl ester of terephthalic acid with ethylene glycol.



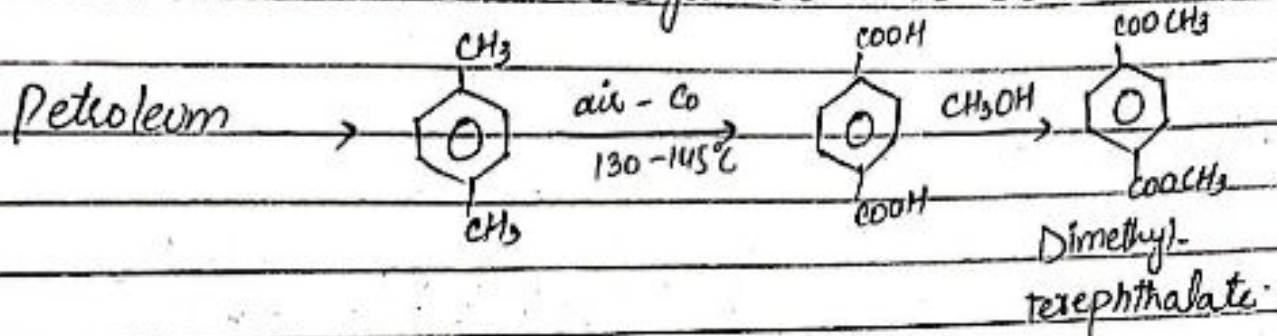
- In many cases, the end groups are hydroxyl ethyl ether $\text{---} \text{CO} \text{---} \text{O} (\text{CH}_2)_2 \text{OH}$.
- The use of ester of acid is preferred on the fact that purity can be achieved easily.
- Ester can be distilled at much lower temp.

Raw material:



TEREPHTHALIC Acid:-

It is prepared by the oxidation of p-xylene obtained from petroleum oxidation is carried out in liq. phase, with air at $130-145^{\circ}\text{C}$ in the presence of an oil soluble cobalt catalyst or with dilute HNO_3 .



Process of manufacturing of Terylene:-

Dimethyl terephthalate (1mole) & ethylene glycol (2mol) in the ratio of 1:6 parts respectively by weight are taken in a reactor connected to vacuum line. The reactors are heated to about $150-160^{\circ}\text{C}$ in the presence of catalyst.

The reaction takes place in two stages
 $\Rightarrow$ low polymers are produced by heating the reactants below 220°C . Methanol is evaporated & condensed.

$\Rightarrow$ These low polymers are condensed together to build up condensed polymer. This stage is carried at 270°C & pressure is reduced to 0.5 torr.

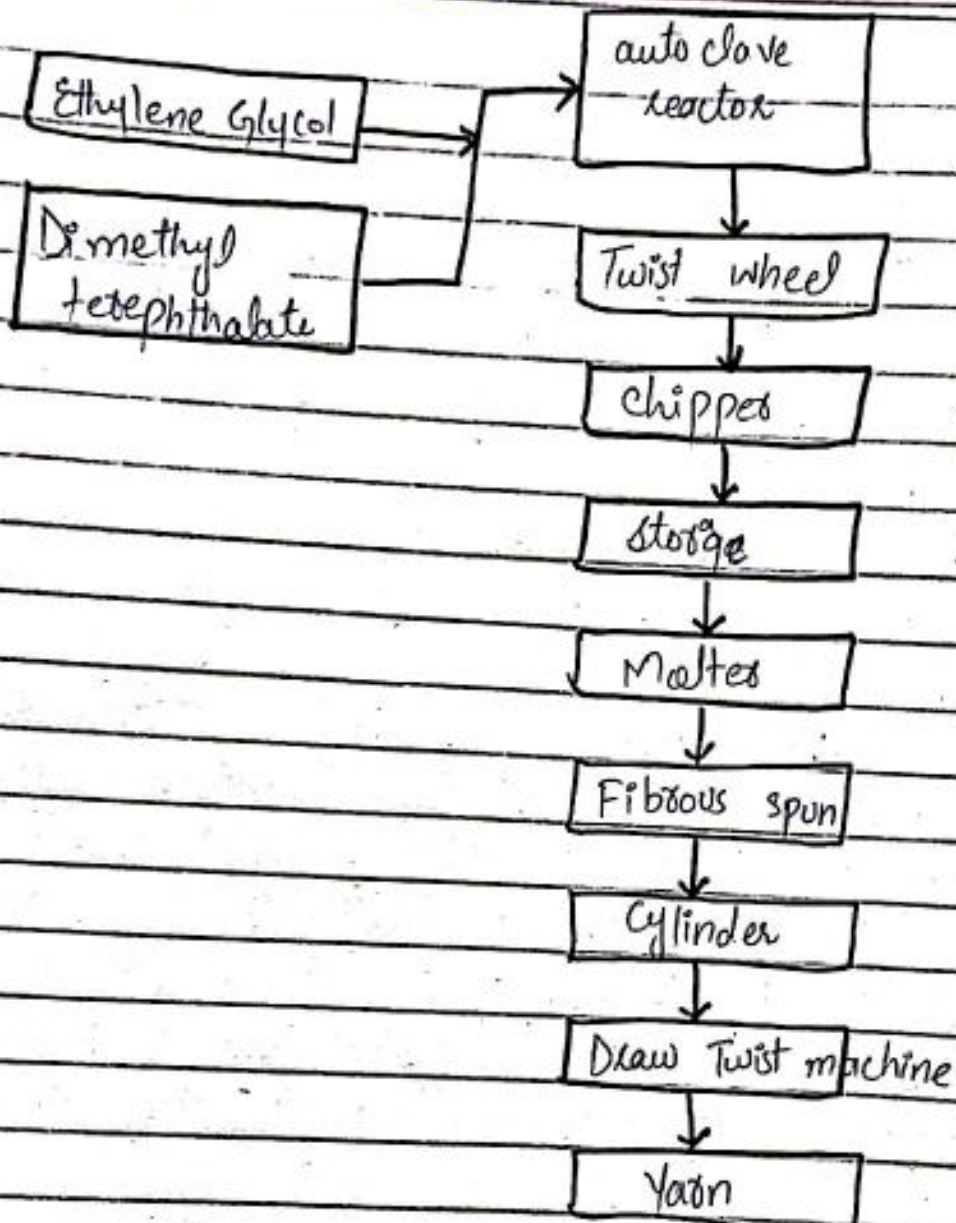
PO_4^{3-} , Sb or Ge compounds are used as catalyst.

The polymer is extruded in the form of ribbon from the ~~an~~ auto clave on a casting wheel. This polymer solidifies on the wheel. The ribbon is cut into pieces & sent to the spinning section where chips are dried to remove moisture and then stored in a hopper for melting.

The fibre is collected through spinnerette from molten polymer. The obtained filaments are solidified instantaneously and are drawn together and wound into cylinders. The yarn is taken to the draw twist machine where it is stretched to about five times of its original length. The diameter reduces 5 times.

Coarse filaments yarns have to be stretched in cold instead of hot. Fine fibres are always stretched in hot. Hot stretching or drawing is always more uniform than cold.

Flow sheet:



Properties of Dacron:

Specific gravity:

1.38

Tenacity:

High (6-7 g/danier) depending upon type of fibre. Normal tenacity is 4.5-5.6 g/danier.

Elasticity:

Good wrinkle resistant due to high elasticity.

Action of alkali & acids:

Strong resistant to both weak acids & bases.

Action of oxidizing agent:

Bleached by oxidizing agents.

Flammability:

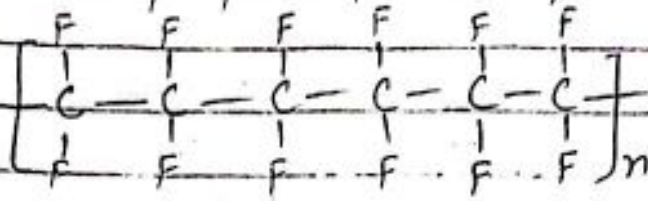
Flammable without spreading fire.

Dyeability:

Can be dyed at 100°C or in the presence of carrier.

~~Teflon~~ (Teflon)

Teflon is polytetrafluoroethylene $(C_2F_4)_n$



It has the composition $CF_2 = CF_2$ which is polymerized.

1) Chemical Resistance:

The fluorine atoms bonded to carbon atoms are unreactive and are packed so densely around the chain of carbon atoms that the carbon atoms are inaccessible & there is no common chemical where molecule can pass the F-atoms to attack C-chain that is why the chemical resistance of ~~teflon~~ teflon is very high.

2) High M.P.:

The uniformity symmetry of molecular teflon chains permit them to pack together in a highly organized way so that the Vander Waal forces b/w the atoms of the adjacent chains produce inter-molecular attractions, so often teflon has very high M.P.

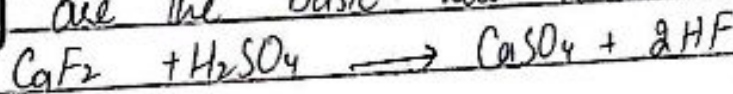
3- High specific gravity:

The molecules are electrically neutral which permit extraordinary close packing which is responsible for high specific gravity of teflon.

Manufacturing process of Teflon

~~Monomer~~: Raw Material

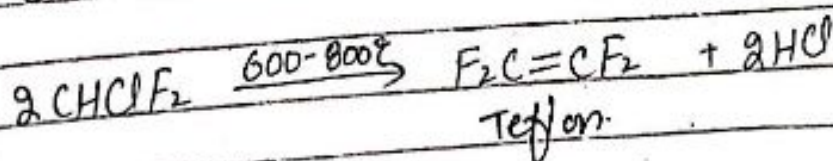
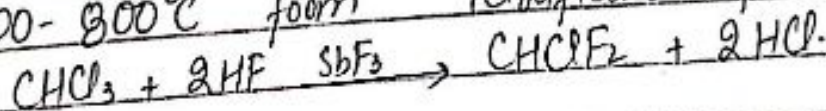
Chloroform and anhydrous hydrogen fluoride [passed by the action of H_2SO_4 on fluor spar (CaF_2)] are the basic raw material.



Manufacturing:

i) Monomer:

Monomer is prepared by the reaction of anhydrous HF with $CHCl_3$ in the presence of SbF_5 to form Difluoro-chloro methane [$CHClF_2$] gas. Its pyrolysis at 600-800°C form Tetrafluoroethylene (Teflon).

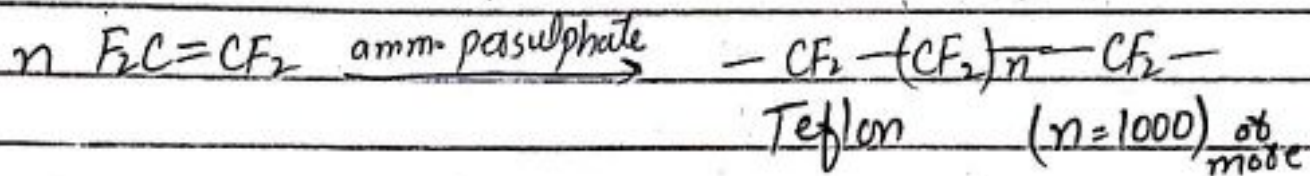


~~Teflon~~ Teflon is scrubbed and recovered.

HCl & other Fluorocarbons are removed which are formed during pyrolysis.

ii- Polymerization:

The purified TFE is then polymerized in stainless steel autoclave in the presence of an aqueous soln. of ammonium persulphate.



The reaction is rapid and exothermic so the conditions must be carefully controlled.

The poly tetrafluoroethylene is a granular white solid which is then washed & dried.

Finishing Process of Cotton

Purification and preliminary process:

The grey cloth, woven cotton fabrics in its loom-state, not only contains impurities, including warp size, but require further treatment in order to develop its full textile potential.

The finishing process for 100% cotton fibers involves a series of stages designed to remove impurities and enhance desirable properties. Further more, it may receive considerable added value by applying one or more finishing processes.

1 Singeing

purpose To burn off loose, protruding fibers (fuzz or hairiness) from the fabric surface.

Process

The cotton fabrics is quickly passed over a hot flame (usually gas flames) or heated plates/cylinders. The fabric process over brushes to raise the fibers then passes over a plate.

Benefits

- creates a smoother, cleaner fabrics surface
- reduce pilling (formation of small balls of fiber)
- improve luster
- provide better base for dyeing and printing

2 Desizing

purpose To remove sizing agents applied to warp yarns before weaving. Sizing agents (like starch, polyvinyl alcohol, or acrylic-based compounds) are added to strengthen the warp yarns and reduce breakage during weaving is called desizing.

Process

In some cases, mixtures of sizing agents of different solubilities are used.

• Oxidative desizing

Oxidative desizing always involves a certain tendering of fibre. For this, break down the sizing material into water soluble components using sodium bromide and ammonium phosphate.

(or) Enzymatic desizing

Enzymatic desizing is often preferred due to its environmental benefits. The solubility and reactivity of enzymes depends upon the Temperature, pH, presence of activators (Na^+ , K^+).

3. Scouring

purpose To remove natural impurities present in raw cotton fabrics, such as waxes, pectins, proteins, oils and other non-cellulosic matter.

Process

Scouring is a chemical washing process.

Fabric is treated with an alkaline solution (sod ash or caustic soda) at an elevated temperature.

This process emulsifies waxes and oils and saponifies fatty acids, making them water-soluble or dispersible, which are then washed away. Surfactants are often added to aid in wetting and dispersion.

Alkali $\Rightarrow$ NaOH & Na_2CO_3

Surfactants $\Rightarrow$ anionic and non-ionic

Cotton materials ^{are taken} in a bath containing agent

i- 2-5% NaOH or Na_2CO_3

ii- 0.5-1.0% anionic or non-ionic surfactants

at 95-100°C for one hour

Benefits

make cotton fabric highly hydrophilic

prepare it for effective bleaching

4 Mercerizing (Spring 2023) S/Q

Purpose

Mercerizing is a textile finishing treatment, primarily applied to cotton, that enhances its properties

Process

Cotton yarn or fabric is immersed under tension in a concentrated solution of caustic soda (NaOH) at a specific temperature (typically $15-20^\circ\text{C}$). The tension is crucial for achieving luster. After treatment, the alkali is washed out and the fabric is neutralized with dilute acid.

Benefits

- (1) Increase luster
- (2) improved strength
- (3) enhanced dye uptake
- (4) improved dimensional stability

Bleaching

Remove the natural yellowish or greyish color of cotton fibers and achieve a desirable level of whiteness. This is essential for bright. Cotton is bleached using oxidizing agents (H_2O_2) in an alkaline bath, often with stabilizers. Sodium hypochlorite bleaching powder can also be used.

Dyeing

Dyeing is a term which refers to dye the textile materials by using many substances.

Printing

The main purpose of textile printing is the production of attractive designs with

well defined boundaries.

Finishing

Treatments given on a textile material to enhance quality after coloration but before the textiles is cut into garments, household textiles or other products.

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